

System Requirements Review

PSYCHE Capstone



*SES*410*

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Team Contributions

- Elysian Hudson: Team Leader
 - Background, Motivation, Requirements
- Caleb Zick
 - Science Question, Requirements
- Daniel Joseph
 - Science Traceability Matrix, Next Steps
- Jacob Coons
 - Science Question, Requirements, Conclusion
- Spencer Emmons
 - Science Traceability Matrix, Requirements



Agenda

- Science Question
- Background
- Motivation
- Requirements
 - Mission-Level Requirements
- Science Traceability Matrix (STM)
- Conclusions
- Next steps



Science Question



- ◆ Determine the feasibility of directly mapping subsurface structures by imparting and recording seismic waves via an artificial impactor and seismograph in order to further the investigations on the origins and evolution of 16 Psyche.



Background



- 16 Psyche is a large asteroid within the main asteroid belt, presumed to be largely composed of metal & hypothesized to be the partial core of a planetesimal.
- Named after the Greek goddess of the soul.
- An orbiter mission to investigate the asteroid is currently underway, scheduled to arrive in 2029.
- Current NASA Psyche mission:
 - Instrument Suite: Multispectral Imager, Gamma-ray and neutron spectrometer, Magnetometer, Gravity science investigation
- Mission precedents: NASA's InSight, Deep Impact, LCROSS missions. JAXA's Hayabusa2 mission.
- Single-site seismology used to develop 3-dimensional maps of subsurface structures of planetary bodies.

Motivation



- Very little is known about the interior structures of small planetary bodies.
- Knowledge about 16 Psyche currently relies on remote sensing observations.
- Directly investigate 16 Psyche to determine if it is the core of a planetesimal, using geophysics and seismology to explore the subsurface.
- Possibly differentiated/layered.
- Learn more about the planetary formation/evolution.
- Future implications for planetary defense and resource exploitation.

Requirements



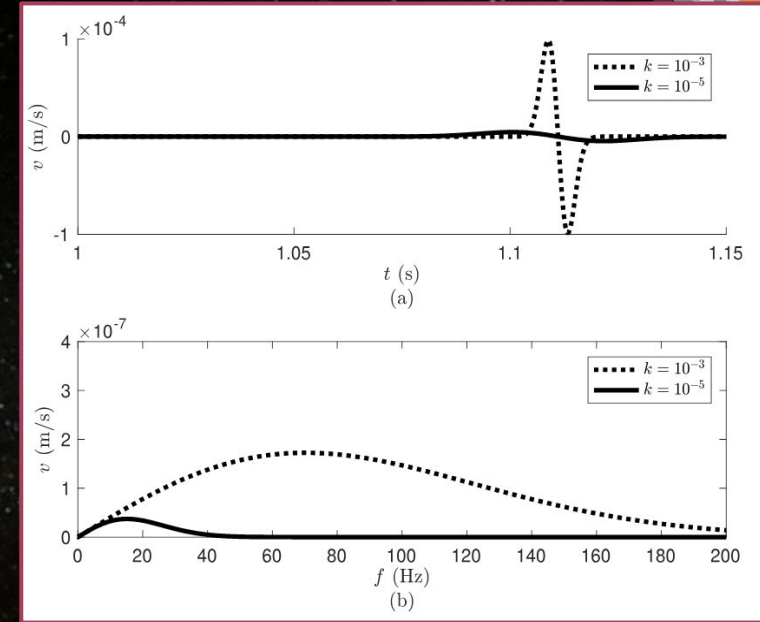
Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
MR1.1	Instrument shall record the propagation of a seismic signal through the asteroid.	TBD	Science Question
MR1.2	Impact device shall generate enough energy to propagate seismic signal throughout the entirety of 16 Psyche.	TBD	Science Question
MR1.3	Lander shall be shielded from debris caused by impact device.	TBD	Science Question

Mission Requirement – 1.1



- Use previously acquired data to estimate frequency range, needed sampling rate, and timing accuracy.
- Use finite element models and seismic wave propagation simulations of Psyche's interior to refine previously estimated data.
- Use simulations to determine optimal seismometer placement.
- Measure changes in seismic signal characteristics as they propagate to map the subsurface interior.



Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.2



- Impact device shall generate enough energy to propagate seismic signal throughout the entirety of 16 Psyche.
- Requires multiple preliminary assumptions that are not yet known to be accurate to 16 Psyche.
- Simulation will be needed to better estimate 16 Psyche's wave velocities, seismic efficiency of impact, $k = E_s / E_k$, and other key values.

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

Where

- F = momentum of impactor
- v_p = longitudinal wave velocity
- v_s = transverse wave velocity
- τ = time constant
- ρ = density of target surface

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.3

- The lander shall be shielded from debris caused by the impact.
- Size, amount, and trajectory of debris will depend on final design of the impactor and the characteristics of the impact site.
- As such, the lander will be located on the far side of 16 Psyche from the impact site so the asteroid itself will provide shielding.



DLR. "Cleanfeed-Animation: Asteroidlander Mascot on Board Hayabusa2." *Vimeo*, 29 Jan. 2025. Clip begins 5:30

Science Traceability Matrix (STM)



Objective	Measurement	Measurement Requirements	Instrument	Instrument Requirements	Data Product
Determine the internal structure/configuration of 16 Psyche by studying the propagation of seismic waves through the interior of the asteroid.	Generate seismic waves that travel through the interior of the asteroid	Collide with enough kinetic energy to produce useful seismic waves	Impact Device	Impact with K.E. on the order of MJ to GJ ($KE = \frac{1}{2} * m * V_{imp}^2$)	Seismic waves within 16 Psyche
	Detect the generated seismic waves	Precision able to detect seismic waves of the generated energy	Seismic Experiment for Interior Structure (SEIS)	Detect the equivalent of a magnitude 1 earthquake from at most 280 km away	Seismogram showing detected waves/psychequake
	Read the characteristics of seismic waves from the seismogram	Differentiate different types of seismic waves (S-Waves, P-Waves, Surface Waves)	Computer Software	Calculate compositions and sizes of 16 Psyche's internal structure	Computer model of 16 Psyche's subsurface structure, layers and differentiation

Conclusions



- This new science question necessitates a second mission to 16 Psyche, and furthers the original science question of 16 Psyche's origins and evolution.
- The requirements for this question necessitate a lander mission, allowing for other science questions to be investigated and activities to be performed that otherwise could not occur in orbit.
- More research is needed!
 - Other potential avenues of interior structural mapping may need to be explored.
 - Kinetic energy requirements may need to be reevaluated.
 - Existing mathematical and programming models may need to be expanded or changed to better approximate 16 Psyche in simulation.

Next Steps



- Follow through on SRR notes and begin preparation for PDR.
- Calculate the energy needed to impart upon 16 Psyche to generate detectable seismic waves.
- Determine device sensitivity required to register seismic waves propagated through 16 Psyche.
- Reach out to more experts in geophysics and seismology regarding further mission development.
- Develop a model to test the calculated energy needed to propagate seismic waves through 280 km of rock and metal to a detectable level by a sensor.
- Using this model, determine the amount of recordings needed to effectively map the subsurface composition of 16 Psyche.

**Thank
You!
Questions?**



Backup Slides:



Requirements Must be SMART:

- Specific -
 - It must address only one aspect of the system design or performance
 - It must be expressed in terms of the need (what and how well), not the solution (how).
- Measurable -
 - Performance is expressed objectively and quantitatively
 - For example, an exact pointing requirement (in degrees) can be tested thus verified prior to launch.
- Achievable -
 - It must be technically achievable at costs considered affordable
 - Note: JWST early designs specified an aperture requirement eventually descoped due to technical issues with deployment.
- Relevant -
 - It must be appropriate for the level being specified
 - For example, requirement on the solar cells should not be designated at the spacecraft level.
- Traceable -
 - Lower level requirements (children) must clearly flow from and support higher level requirements (parents).
 - Requirements without a parent are referred to as orphans, and need to be assessed for necessity of inclusion.

STM Contents

(Groppi)

- Teams developing NASA missions use an STM to summarize the science and top-level requirements.
- An STM has the following columns:
 - **Science Objectives:** The elements of your science question(s)
 - **Measurement Objectives:** What quantities do you need to measure to achieve the science objective:
 - **Science Measurement Requirements:** What do you need to measure and how well do you need to measure it? Expected measurement range and required measurement error.
 - **Instrument Functional Requirements:** What other requirements does the instrument that takes the data need to have to meet your needs?
 - **System Functional Requirements:** What other requirements does the whole experiment need to have to meet your requirements?
 - **Science Product:** What is the data your experiment will produce (the low-level data measured, not the answer you get after you analyze it).